✦ TECHNICAL WORKSHOP

Intro to RAG

EdinburghAI | Retrieval-Augmented Generation & Agentic Systems

SEMANTIC EMBEDDINGS
High-dimensional vector representations

VECTOR RETRIEVAL
Cosine similarity & Top-K chunking

AGENTIC WORKFLOWS
Autonomous routing & self-reflection

The Motivation for RAG



THE CORE PROBLEM

"LLMs make fake claims but they sound confident whilst doing so."

Why Hallucination Happens:

- Parametric memory contains gaps and outdated training knowledge.
- Generative models optimize for fluent syntax, not empirical truth.

Solution: Ground answers in retrieved source documents before generation.

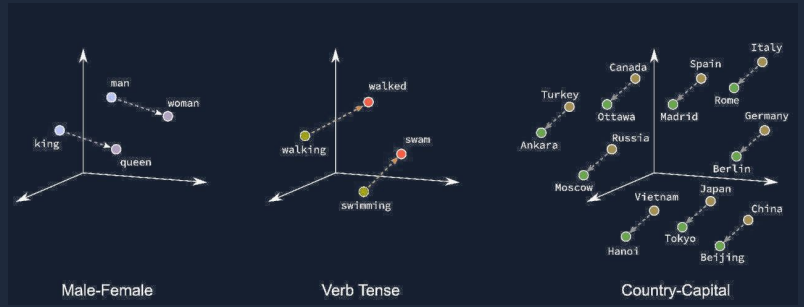
Background: Embeddings

KEY CONCEPT: Represent the meaning of text numerically, such that texts with similar meaning have similar embeddings.

2D Semantic Space (Clusters by Domain)



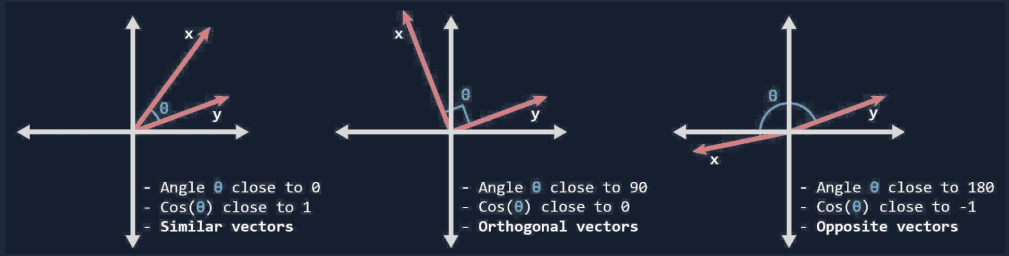
High-Dimensional Vector Arithmetic (Analogies)



Background: Measuring Similarity

LINEAR ALGEBRA FOUNDATION: Embeddings are just vectors, so we use vector angles to measure semantic similarity.

Vector Geometric Relationship & Cosine Metric



$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}$$

SIMILARITY INTERPRETATION

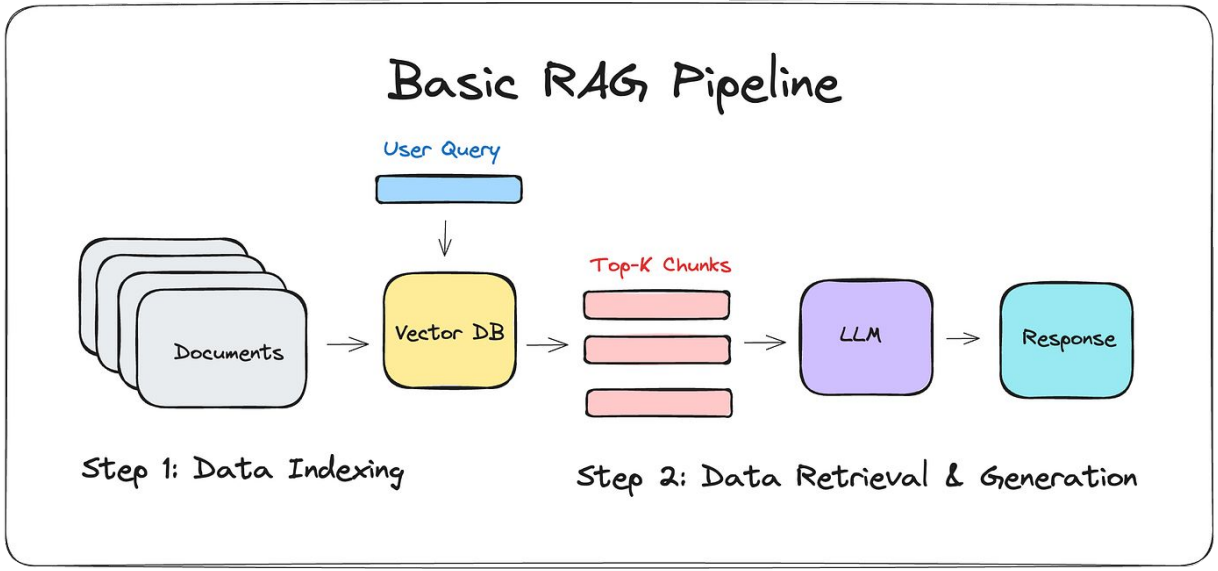
Cos(θ) ≈ 1.0 (Angle ~ 0°)
Identical / highly aligned semantics.

Cos(θ) ≈ 0.0 (Angle ~ 90°)
Orthogonal; unrelated semantic topics.

Cos(θ) ≈ -1.0 (Angle ~ 180°)
Diametrically opposite meanings.

Retrieval Augmented Generation (RAG)

CORE ARCHITECTURE: Provide the LLM with relevant document chunks, then instruct it to synthesize answers informed by these chunks.



The 5-Step Retrieval Pipeline

WORKFLOW: How documents are ingested, transformed into semantic vectors, and matched against search queries.

01

Corpus Ingestion

First we need a collection of raw documents forming the knowledge base.

02

Chunking

Chunk documents into smaller, coherent segments for retrieval.

03

Segment Embeddings

Produce numerical embeddings representing the meaning of each chunk.

04

Query Embedding

Embed the user's incoming search query into the same vector space.

05

Vector Matching

Compute similarity and return the chunks closest to the search query.

Limitations of Basic RAG

SYSTEMIC BOTTLENECKS: Standard naive RAG pipelines suffer from single-pass vulnerability and absence of query reflection.

LIMITATION 01

Single Retrieval Step

Only one retrieval step—if initial retrieved chunks are poor or noisy, the final response will inevitably fail.

LIMITATION 02

Query-Document Mismatch

Static similarity has no validation mechanism to evaluate whether retrieved chunks truly answer the user's intent.

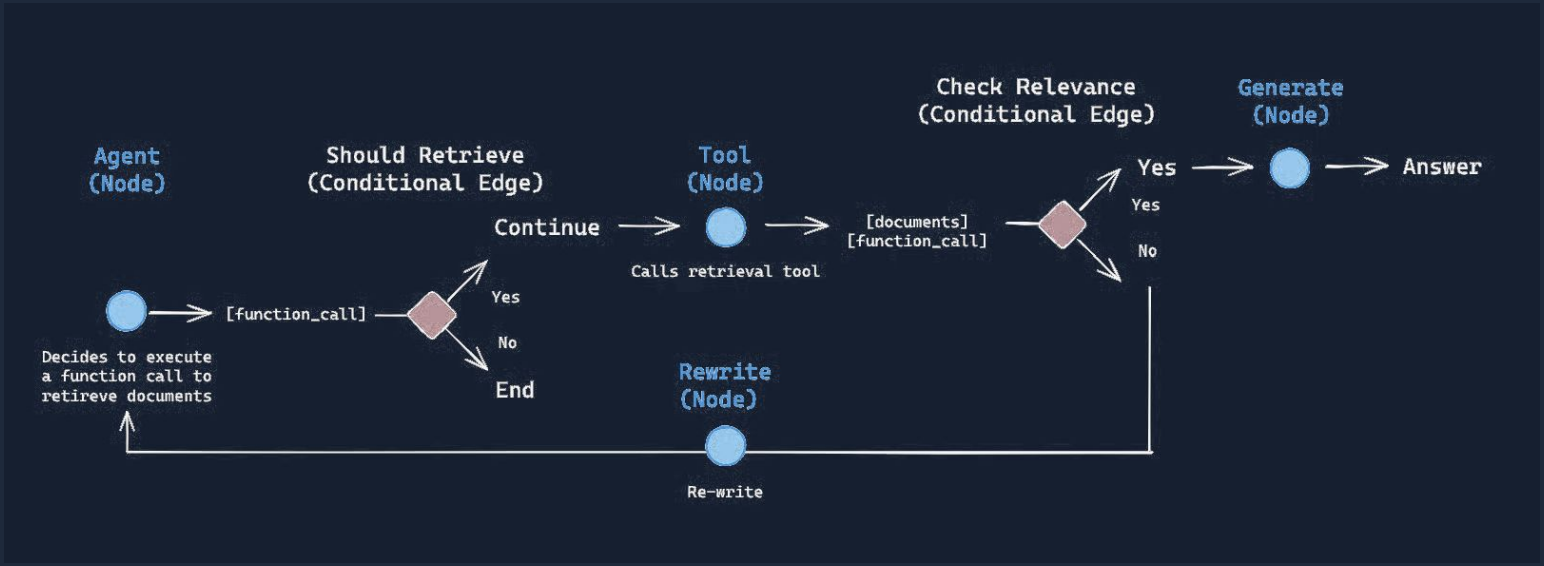
LIMITATION 03

No Multi-Step Reasoning

Limited reasoning capabilities: does not allow for multi-hop decomposition, iterative refinement, or synthesis.

Agentic RAG Architecture

DEFINITION: An autonomous LLM-powered agent with dynamic tool use, conditional routing, and iterative reflection.



Main Benefits of Agentic RAG

STRATEGIC ADVANTAGES: Elevating simple semantic search into an adaptive, self-correcting intelligence loop.

BENEFIT 01

Query Transformation

The agent can rephrase and decompose ambiguous user prompts into multiple retrieval-friendly search queries.

BENEFIT 02

Multi-Pass Retrieval

Performs multiple sequential or parallel retrievals across diverse corpora until sufficient context is gathered.

BENEFIT 03

Contextual Relevance Reasoning

The agent reasons over retrieved documents, scoring relevance and filtering noise before generation.

BENEFIT 04

Self-Critique & Active Refinement

Validates candidate answers against retrieved evidence, triggering automated re-querying when gaps remain.